

Summary

Full-stack engineer who ships AI-backed products end to end. Built the retrieval-grounded chatbot behind [Small Business PEAK](#), an Australian Government-funded service that national media reported as one of the first government-backed uses of an AI chatbot to explain regulatory change. Runs a live paid AI product with its own Stripe purchase-to-delivery loop, and is building a [Python retrieval and voice-agent platform](#) with hybrid dense/sparse retrieval and cross-encoder reranking. Five years of production delivery across consulting, SaaS and a production learning platform.

5 yrs	20+	30.5k	Gov't
Production delivery	Client systems delivered	Python lines, 60k tests	Funded AI chatbot, in media

Core Skills

- AI & LLM:** retrieval-augmented generation, embeddings, hybrid dense/sparse retrieval, reciprocal rank fusion, cross-encoder reranking, grounding and scope limits, prompt engineering, structured outputs, Amazon Bedrock, OpenAI and Anthropic APIs, LiveKit voice agents.
- Python & backend:** Python 3.12, FastAPI, Pydantic v2, psycpg 3, PostgreSQL, queue-based workers, asynchronous processing, provider abstraction, pytest, ruff, mypy.
- Full-stack & web:** TypeScript, React, Next.js, Node.js, PHP/Laravel, SQL, MySQL, Redis, REST APIs, webhooks, accessibility-aware and SEO-aware delivery.
- Delivery & operations:** Docker, GitHub Actions CI, AWS, Cloudflare, Vercel, Supabase, Stripe, Git, Linux, OpenTelemetry, structured logging.
- Client & product:** direct stakeholder scoping, requirement translation, end-to-end ownership across 20+ concurrent engagements, analytics instrumentation, iteration on live feedback.

Work Experience

Independent Developer - AI Platform & Web Projects Canberra 11/2025-Present

- ROLE** Full-time independent engineer running a live paid SaaS product while building a [Python retrieval and voice-agent platform](#).
- ACTIONS** Own the full purchase-to-delivery loop for PicPolisher, including Stripe webhook handling designed so a retry or an out-of-order event cannot lose or duplicate an order. Building a companion Python service that ingests documents and websites, retrieves and reranks answers over them, plus a voice-agent prototype on LiveKit Agents.
- IMPACT** Grew the Python platform to roughly 30,500 lines of source across 140 modules, backed by roughly 60,000 lines of tests; PicPolisher remains live and processing real customer purchases.

Software Engineer - Acorn PLMS Canberra 05/2025-10/2025

- ROLE** Built and maintained a core capability-evaluation module inside an established production learning platform.
- ACTIONS** Integrated [Amazon Bedrock](#) into existing workflows inside established permission boundaries, then replaced cron-heavy processing with [queue-based jobs and caching](#).
- IMPACT** Removed N+1 query patterns and improved production reliability; radar-chart reporting views drew positive feedback from clients, colleagues and a supervisor.

Full-stack Developer - OPF Consulting Canberra 05/2023-05/2025

- AI DELIVERY** Built the AI chatbot for [Small Business PEAK](#) (smallbusinesspeak.org.au), a COSBOA service funded by the Department of Employment and Workplace Relations. It answers small-business questions on workplace relations law, grounded in a curated corpus through embeddings, retrieval and reranking so answers stay inside source material.
- CLIENTS** Delivered and maintained [more than 20 client websites and digital systems](#), running scoping calls directly with non-technical stakeholders and translating requirements into production builds across many concurrent engagements.
- IMPACT** Australian media reported the Small Business PEAK launch as one of the first government-backed uses of an AI chatbot to explain regulatory change; the service remains live.

- Frontend Developer - Prudential Technology

Shenzhen, China

10/2022-05/2023
- ACTIONS

Built a cross-platform lifestyle application from scratch on React, React Native and TypeScript, backed by Google Cloud and Firebase.
- Full-stack Developer - Tencent

Shenzhen, China

11/2021-09/2022
- ACTIONS

Built React and Ant Design interfaces for a user-segmentation and marketing platform, plus Node/Egg and MySQL service-layer features between those interfaces and back-end systems.
- IMPACT

Improved interaction quality and maintainability while the team's work contributed to 3x user growth.

Projects

AI & delivery evidence

- hybrid-rag-service - Python retrieval & voice-agent platform

IN DEVELOPMENT
- Python 3.12, FastAPI, Pydantic v2, pycopg 3, PostgreSQL, Docling, crawl4ai, LiveKit Agents, Docker
- BUILT

A multi-tenant knowledge service that ingests documents and websites and answers questions over them, using hybrid dense/sparse retrieval, [reciprocal rank fusion](#) and cross-encoder reranking with calibrated confidence thresholds and an explicit degraded mode, plus a companion [voice-agent service on LiveKit Agents](#) with its own evaluation harness.
- DECISIONS

Kept the deployment footprint small - no separate vector database, no LangChain, no Kubernetes - so the service can be stood up inside a customer-controlled environment; queue-based workers handle document and website intake asynchronously.
- PicPolisher - Paid AI product

LIVE
- Next.js, TypeScript, Supabase/PostgreSQL, Stripe, Vercel
- OPERATE

The full path from purchase to delivery - authenticated customer flows, Stripe webhook events, database records and asynchronous generation - with explicit states so a retry or out-of-order webhook cannot lose or duplicate an order. [picpolisher.com](#)
- OWNED

Product planning, engineering, deployment, analytics and iteration, as sole engineer.
- Health-service websites

CLIENT WORK
- Next.js, TypeScript, AI-assisted delivery
- DELIVERED

Production clinic websites including [hhanda.com.au](#) for Her Health & Aesthetics and [neurodiversityninjas.com.au](#), covering service taxonomy, booking pathways, accessibility and launch.
- METHOD

Built AI-assisted end to end - generated click-through HTML wireframes for client sign-off before implementation, then reviewed every generated change against accessibility and content-structure standards before it shipped.

Education

- Central China Normal University - Master of Journalism

09/2014-06/2017
- Central China Normal University - Double Bachelor of Information Technology and Journalism

09/2010-06/2014

Languages

English: professional fluency | [Mandarin Chinese](#): native